PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (http://bmjopen.bmj.com/site/about/resources/checklist.pdf) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Spirometry parameters used to define small airways obstruction in population-based studies: Systematic review protocol
AUTHORS	Knox-Brown, Ben; Mulhern, Octavia; Amaral, Andre F S

VERSION 1 – REVIEW

REVIEWER	Corriveau, Sophie McMaster University
REVIEW RETURNED	12-Jul-2021

GENERAL COMMENTS	There are no specific comments to the author.

REVIEWER	Corradi, Massimo University of Parma, Department of Clinical Medicine, Nephrology and Health Sciences.
REVIEW RETURNED	16-Aug-2021

GENERAL COMMENTS	Authors described a protocol they are going to apply to better define SAO. The topic is relevant, but I do not think that a protocol is sufficient for a publication

REVIEWER	Welch, Lindsay University of Southampton, School of Health Sciences, School of Health Sciences
REVIEW RETURNED	27-Aug-2021

GENERAL COMMENTS	BMJ Open review Systematic review of spiro parameters in SAO. Protocol Abstract. Good summary of the context, and potential of distal flow rates to predict future airways disease. I agree this is an important area in which to collate the evidence to define clear parameters. Methods Pleased to see narrative analysis and meta-analysis being conducted here. However, for a systematic review I would suggest

	that more databases are included; PubMED, Conchrane, NHS libraries, and others. Background A well drafted explanation of the multiple methods that can be used to calculate or define the parameters of measuring small airways. Very clear aim and pleased to see the inclusion here of primary and secondary research questions. Key research question is to define the parameters for population-based studies; also, of interest would be the possibility of prognostication. Methods and search strategy Again my main comment here would be to expand the data bases – state this as a global search (noted all languages are included) but be explicit about the systematic nature of the searching, and include: CINAHL, Cochrane Library, EMBASE, Google Scholar, MEDLINE, Pub Med, SCOPUS, Web of Science, NICE Website, NHS England Website, NIHR Library, Europe PMC Study records I am not familiar with COVIDENCE software – so I am pleased you added the link. It appears to be a robust and sensible data management system. Well described sections on quality assessment and the prioritisation of outcomes. Discussion My main question whilst reviewing this protocol, was, why are you publishing the protocol, does this does not detract from the publication of the review itself. However, you address this robustly in the discussion section. I concur that it adds rigour and the opportunity for others to repeat the review to pool statistically accurate spirometric parameters in SAO. Dissemination I would suggest here that you like this work to guideline committees, or the BTS or Thorax, to maximise clinical dissemination and adoption. Furthermore, you could make suggestions of how you envisage these parameters impacting on clinical practice. How will you ensure that if you find clear parameters – these are 1- Evaluated in a population-based study 2- Plan to be presented to guidelines committee

VERSION 1 – AUTHOR RESPONSE

Response to reviewer 3: Ms. Lindsay Welch

Thank you for taking the time to comment on our protocol, please see below our responses.

Methods

Pleased to see narrative analysis and meta-analysis being conducted here. However, for a systematic review I would suggest that more databases are included; PubMED, Conchrane, NHS libraries, and others.

We chose 2 large databases, PubMed and Web of Science (WOS), and used Google scholar to look for grey literature. We restricted our search to these databases for several reasons: 1) PubMed and WOS are two of the largest databases available and, on initial testing, they produced a similar number of search results, with some duplicate articles. This suggests that our search strategy is effective covering both clinical/biomedical literature (PubMed) and interdisciplinary literature (WOS); and 2) the search of grey literature using Google scholar is robust and shown to be very effective at capturing studies that may be missed by larger databases[1]. In addition, our search strategy includes looking at the reference lists of relevant studies to identify any studies that would have been missed by our search strategy. Cochrane and NHS libraries are in patient populations so not relevant to our review in the general population.

Background

A well drafted explanation of the multiple methods that can be used to calculate or define the parameters of measuring small airways. Very clear aim and pleased to see the inclusion here of primary and secondary research questions. Key research question is to define the parameters for population-based studies; also, of interest would be the possibility of prognostication.

We did originally think about looking at prognostication, however when we did initial searches there is very little data in population-based studies. The majority of studies looking at prognostication are in clinical populations (e.g. alpha-1-antitrypsin deficiency, COPD, heart failure and asthma), which did not fit the intended aim of our review of studies based in the general population. Prognostication in clinical populations is highly relevant, but we feel it should be the focus of its own review.

Methods and search strategy

Again my main comment here would be to expand the data bases – state this as a global search (noted all languages are included) but be explicit about the systematic nature of the searching, and include: CINAHL, Cochrane Library, EMBASE, Google Scholar, MEDLINE, Pub Med, SCOPUS, Web of Science, NICE Website, NHS England Website, NIHR Library, Europe PMC Study records I am not familiar with COVIDENCE software – so I am pleased you added the link. It appears to be a robust and sensible data management system. Well described sections on quality assessment and the prioritisation of outcomes.

See above for rationale regarding search strategy and database choices.

Discussion

My main question whilst reviewing this protocol, was, why are you publishing the protocol, does this does not detract from the publication of the review itself. However, you address this robustly in the discussion section. I concur that it adds rigour and the opportunity for others to repeat the review to pool statistically accurate spirometric parameters in SAO.

The original intention behind submitting the protocol was to first and for most be transparent in our approach, but also to follow a practice that is increasingly being adopted by journals. An increasing number of journals are now asking that study protocols are published in peer reviewed journals.

Dissemination

I would suggest here that you like this work to guideline committees, or the BTS or Thorax, to maximise clinical dissemination and adoption. Furthermore, you could make suggestions of how you envisage these parameters impacting on clinical practice. How will you ensure that if you find clear parameters – these are 1- Evaluated in a population-based study 2- Plan to be presented to guidelines committee.

Thank you for your suggestion. We will disseminate the findings of this review as wide as possible via both a peer-reviewed respiratory journal and scientific fora. Our intention is to build upon the findings of this review and conduct further research on this topic in a large population-based study. After this is completed, we could think about engaging with a guidelines committee.

VERSION 2 – REVIEW

REVIEWER	Welch, Lindsay University of Southampton, School of Health Sciences, School of Health Sciences
REVIEW RETURNED	15-Sep-2021

GENERAL COMMENTS	Dear Author Thank you for your revision. I note only minor alterations marked up in your resubmitted manuscript; which was of high quality from the outset. However, I do recommend that you include a wider array of databases for your systematic review. Many thanks

VERSION 2 – AUTHOR RESPONSE

R1. We appreciate the suggestion to use other databases in addition to Medline (PubMed), Web of Science and Google Scholar, which are already included in our search strategy, to find more studies for our review. However, due to time constraints, we cannot do it now.